- Proof transcript

The verifier program must output:

- ACCEPT if the proof is valid
- REJECT otherwise

6.7 Security Requirements

The implemented protocol must satisfy:

- **Completeness:** Honest provers with valid secret shares can convince the verifier.
- **Soundness:** A prover without knowledge of both secret shares cannot convince the verifier.
- **Zero-Knowledge:** The verifier learns no information about the secret values.

6.8 Implementation Constraints

Students must:

- Implement all proof protocols from scratch.
- Implement both interactive and non-interactive versions.
- Use only standard hash functions (e.g., SHA-256).
- Not use any external cryptographic protocol or zero-knowledge proof libraries.

Problem 7: Zero-Knowledge Proof of Correct Linear Model Inference with Binary Inputs

7.1 Problem Description

Machine learning models are often used on private input data. In privacy-sensitive applications, it is desirable to prove that a model output was computed correctly without revealing the private input itself.

In this assignment, a public linear model is defined by a weight vector and bias term:

$$W = (w_1, w_2, \ldots, w_n), \quad b \in \mathbb{Z}_q$$

The prover possesses a private input vector:

$$x = (x_1, x_2, \ldots, x_n)$$

where each input satisfies the constraint:

$$x_i \in \{0, 1\}$$

The prover computes the model output:

$$y = \sum_{i=1}^{n} w_i x_i + b \pmod{q}$$

The prover must convince the verifier that:

- The committed input values are binary (each value is either 0 or 1)
- The output y has been computed correctly using the public model parameters and the private input vector

without revealing any information about the private input vector.
This must be achieved using a Zero-Knowledge Proof protocol.

7.2 Public Inputs

The following information is publicly known to both the prover and the verifier:

- Group parameters (G, q, g, h) for the commitment scheme
- Model weight vector:
$$W = (w_1, w_2, \ldots, w_n)$$
- Bias term:
$$b \in \mathbb{Z}_q$$
- Commitments to input values:
$$C_1, C_2, \ldots, C_n$$
where each commitment is of the form:
$$C_i = g^{x_i} h^{r_i}$$
- Claimed model output:
$$y \in \mathbb{Z}_q$$
- Number of protocol rounds k

These values will be provided to students as part of the assignment input files.

7.3 Private Inputs

The prover possesses the following secret information:

- Private input vector:
$$x_1, x_2, \ldots, x_n \in \{0, 1\}$$
- Commitment randomness:
$$r_1, r_2, \ldots, r_n \in \mathbb{Z}_q$$

such that:

$$C_i = g^{x_i} h^{r_i}, \quad \forall i$$

and

$$y = \sum_{i=1}^{n} w_i x_i + b \pmod{q}$$

7.4 Goal

The prover must convince the verifier of the following statement:

"I know input values $x_1, x_2, \ldots, x_n$ and randomness values corresponding to commitments $C_1, C_2, \ldots, C_n$, such that each input is binary and the model output y has been computed correctly."

while ensuring that:

- The verifier learns no information about the input vector.
- The verifier cannot determine any individual input value.
- A prover who does not know valid input values cannot convince the verifier.

7.5 Protocol Requirements

Students must implement the following components:

1. **Commitment Generation**

   Implement the Pedersen commitment scheme and generate commitments to each input value.

2. **Interactive Zero-Knowledge Protocol**

   Implement a multi-round interactive zero-knowledge proof protocol that allows the prover to demonstrate:

   - Correct commitment to input values
   - Satisfaction of binary constraints
   - Correct computation of the model output

3. **Non-Interactive Version**

   Implement a non-interactive version of the protocol using the Fiat–Shamir transform, where challenges are generated using a cryptographic hash function.

4. **Prover Program**

   The prover program must take as input:

   - Private input vector
   - Commitment randomness
   - Model parameters

   and must output:

   - Commitment values
   - Claimed output value
   - Proof transcript

5. **Verifier Program**

   The verifier program must take as input:

- Commitment values
- Model parameters
- Claimed output value
- Proof transcript

and must output:

- ACCEPT if the proof is valid
- REJECT otherwise

7.6 Input and Output Specifications

Students will be provided with:

- Group parameters (G, q, g, h)
- Model parameters (W, b)
- Protocol configuration parameters

Students must implement both prover and verifier programs.

7.7 Security Requirements

The implemented protocol must satisfy:

- **Completeness:** An honest prover can convince the verifier.
- **Soundness:** A prover without valid input values cannot convince the verifier.
- **Zero-Knowledge:** The verifier learns no information about the private input vector.

7.8 Implementation Constraints

Students must:

- Implement all commitment schemes and proof protocols from scratch.
- Implement both interactive and non-interactive versions.
- Use only standard hash functions (e.g., SHA-256).
- Not use any external cryptographic protocol or zero-knowledge proof libraries.

Submission Instructions

Submit via the course submission portal.

Each problem must include:

- Source code
- Execution instructions
- Report

End of Assignment